

# Prepare Abiotic Data for Model Fitting

## Description

Reshapes long-format abiotic data into a wide data frame suitable for downstream scaling, filtering to the canonical sample index supplied by 'align\_sample\_ids()'.

## Usage

```
prepare_abiotic_for_fit(data_abiotic_long = NULL, data_sample_ids = NULL)
```

## Arguments

`data_abiotic_long` A data frame in long format with columns 'dataset\_name', 'age', 'abiotic\_variable\_name', and 'abiotic\_value'.

`data_sample_ids` A data frame of valid '(dataset\_name, age)' pairs as returned by 'align\_sample\_ids()'.

## Details

Unlike the former 'prepare\_data\_for\_fit(type = "abiotic")', this function retains 'dataset\_name' and 'age' as real columns so that 'scale\_abiotic\_for\_fit()' can operate on them without parsing row names. Row names are added by 'scale\_abiotic\_for\_fit()' as part of the scaling step.

## Value

A data frame in wide format with real columns 'dataset\_name', 'age', and one column per abiotic variable. Rows are ordered by 'dataset\_name' then 'age', matching the order in 'data\_sample\_ids'. Missing variable-sample combinations are left as 'NA'. The 'age' column is preserved as a plain numeric column (not encoded in row names) to simplify downstream scaling.

## See Also

[align\_sample\_ids()], [scale\_abiotic\_for\_fit()], [build\_jsdm\_fit\_input()]